Skills Summary

Diagnosing Invalid Simplification of Sums of Radicals

Use this reference while completing your worksheet or when studying.

Skill 1 — Simplifying One Radical

Roots split over products, never over sums: $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$, and there is no rule for $\sqrt{a+b}$.

- Look for the largest perfect-square factor: $4, 9, 16, 25, 36, 49, 64, 81, 100$.
- Take its root out front and leave the rest under the sign.
- A radicand with no perfect-square factor is already simplest.

Example: Write $\sqrt{45}$ in simplest radical form.
Step 1. Hunt for the largest perfect square that divides the radicand, because only a product may be split.
Step 2. $45 = 9 \cdot 5$, so $\sqrt{45} = \sqrt{9}\,\sqrt{5} = \mathbf{3\sqrt{5}}$.

Try it: Write $\sqrt{98}$ in simplest radical form.

check: $\mathbf{7\sqrt{2}}$

Skill 2 — Combining Like Radicals

Radicals combine like terms: $2\sqrt{2} + 3\sqrt{2} = 5\sqrt{2}$, exactly as $2x + 3x = 5x$.

- Simplify every term *first* — unlike radicands often match once simplified.
- Add only the coefficients. The radicand is never added and never changes.
- If the radicands still differ after simplifying, the expression is finished as it stands.

Example: Simplify $\sqrt{20} + \sqrt{45}$.
Step 1. The radicands look unlike, so simplify each term before deciding whether anything combines.
Step 2. $\sqrt{20} = 2\sqrt{5}$ and $\sqrt{45} = 3\sqrt{5}$, so the coefficients add to give $\mathbf{5\sqrt{5}}$.

Try it: Simplify $\sqrt{75} - \sqrt{12}$.

check: $\mathbf{3\sqrt{3}}$

Skill 3 — Testing a Claim with Decimals

When a classmate's rule looks plausible, price it out. Put a decimal on both sides; if the two numbers differ, the rule is dead.

- Perfect squares make the sharpest counterexamples — no rounding is involved at all.
- A decimal test disproves a rule but never proves one; state the answer in exact radical form.

Example: Is $\sqrt{4} + \sqrt{9}$ equal to $\sqrt{13}$?
Step 1. Both radicands are perfect squares, so each root can be taken exactly and the claim needs no rounding to settle.
Step 2. The left side is $2 + 3$, while $\sqrt{13}$ is about 3.61, so $\mathbf{5 > \sqrt{13}}$.

Try it: Compare $\sqrt{1} + \sqrt{9}$ with $\sqrt{10}$.

check: $\mathbf{4 > \sqrt{10}}$